DAY 22 – SECURITY LOG FUNDAMENTALS

Cyber Security Internship – Veda Technology

1. Objective

To understand the fundamentals of security logs, study common security events, and identify important log fields such as timestamps, usernames, source IP addresses, and event types. The task focuses on analysing authentication logs to recognize suspicious activities and understand basic security event monitoring.

2. Introduction

Security logs are records generated by operating systems, applications, servers, and network devices to track activities and events.

These logs provide information about login attempts, authentication failures, user activities, system errors, and potential security threats.

Security log analysis is an essential cybersecurity practice because it helps security professionals identify unusual behavior, investigate incidents, and monitor system activities.

Linux systems maintain different types of logs that help administrators understand system operations and security-related events.

Common Types of Security Logs

Log Type	Description
Authentication Logs	Record user login attempts and authentication events
System Logs	Record operating system activities and errors
Firewall Logs	Record permitted or blocked network traffic
Application Logs	Record application activities and errors
Audit Logs	Track security-related system events and user actions

3. Important Security Log Fields

Security logs contain several important fields that help analysts investigate events.

Field	Description	Example
Timestamp	Date and time of an event	2026-10-22 10:15:30
Username	User associated with the event	admin
Source IP	IP address associated with the event	192.0.2.10
Event Type	Category of activity	Failed Login
Status	Result of the activity	Success / Failure

Service	System service generating the log	sshd

Importance of Log Fields

- Timestamp: Helps determine when an event occurred and establish an incident timeline.
- Username: Identifies the account associated with the activity.
- Source IP: Helps determine the reported network origin of an event.
- Event Type: Indicates the action recorded by the system.
- Status: Shows whether the activity succeeded or failed.

4. Practical Implementation

The task demonstrates basic security log analysis using synthetic Linux authentication logs.

Step 1: Create a Sample Authentication Log

A sample file named auth_sample.log is created using a Linux text editor.

Command:

nano auth_sample.log

Sample Log Data:

2026-10-22 10:15:30 sshd Failed password for admin from 192.0.2.10

2026-10-22 10:16:12 sshd Failed password for admin from 192.0.2.10

2026-10-22 10:17:05 sshd Failed password for admin from 192.0.2.10

2026-10-22 10:20:15 sshd Accepted password for yuktha from 192.0.2.20

2026-10-22 10:25:40 sshd Failed password for root from 198.51.100.15

2026-10-22 10:30:10 sshd Accepted password for intern1 from 192.0.2.30

Description:

The synthetic log contains six authentication events, including successful and failed login attempts.

The entries are designed to demonstrate how security analysts examine authentication activity and identify suspicious patterns.

Step 2: Display the Log File

Command:

cat auth_sample.log

Description:

The cat command displays the contents of the log file in the Linux terminal.

It helps analysts review recorded authentication events.

Step 3: Identify Failed Login Attempts

Command:

grep "Failed password" auth_sample.log

Description:

The grep command searches the log file for authentication failures.

Observation:

Four failed login attempts are present in the synthetic dataset.

Repeated authentication failures may indicate incorrect credentials, unauthorized access attempts, or password-guessing activity.

Step 4: Identify Successful Login Attempts

Command:

grep "Accepted password" auth_sample.log

Description:

This command filters successful authentication events.

Observation:

Two successful login events are recorded for users yuktha and intern1.

Successful logins should also be reviewed to ensure that the access was authorized.

Step 5: Identify Activity from a Specific IP Address

Command:

grep "192.0.2.10" auth_sample.log

Description:

This command filters authentication events associated with the specified source IP address.

Observation:

The IP address 192.0.2.10 appears in three failed login attempts within approximately two minutes.

This pattern requires further investigation because it may indicate repeated password-guessing attempts.

5. Security Log Analysis

The sample authentication log was examined to identify event patterns and possible security concerns.

Time	Username	Source IP	Event	Status
10:15:30	admin	192.0.2.10	Login Attempt	Failed
10:16:12	admin	192.0.2.10	Login Attempt	Failed
10:17:05	admin	192.0.2.10	Login Attempt	Failed
10:20:15	yuktha	192.0.2.20	Login Attempt	Success
10:25:40	root	198.51.100.15	Login Attempt	Failed
10:30:10	intern1	192.0.2.30	Login Attempt	Success

Event Summary

Parameter	Count
Total Events	6
Successful Logins	2
Failed Logins	4
Unique Source IPs	4
Unique Usernames	4
Repeated Failures from One IP	3

Security Observations

1. Three consecutive failed login attempts were recorded for the admin account from the same source IP.
2. A failed authentication attempt targeted the root account.
3. Two successful login events were recorded for separate user accounts.
4. Repeated failed attempts may suggest password guessing, but additional evidence is required to confirm malicious activity.
5. The source IP addresses are documentation ranges used for simulation and do not represent actual attackers.

6. Potential Security Threats Identified

Security Concern	Description	Recommended Action
Repeated Login Failures	Multiple unsuccessful attempts against one account	Investigate and configure login rate limiting

Root Login Attempt	Authentication attempt targeting the root account	Restrict direct root login
Unauthorized Access	Login from an unexpected source	Verify user activity and source
Password Guessing	Repeated attempts to discover credentials	Enable MFA where supported
Suspicious Login Pattern	Unusual authentication activity	Configure monitoring and alerts

7. Importance of Security Log Analysis

Security log analysis plays an important role in protecting computer systems and networks.

- Threat Detection: Helps identify suspicious login attempts and abnormal system activities.
- Incident Investigation: Provides evidence for examining security incidents.
- User Activity Monitoring: Helps administrators review authentication and account usage.
- System Troubleshooting: Assists in identifying errors and operational problems.
- Security Auditing: Supports compliance checks and access reviews.
- Early Warning: Enables detection of potential attacks before they escalate.

8. Best Practices for Security Log Management

1. Enable logging for important system and authentication events.
2. Regularly monitor failed login attempts and suspicious activities.
3. Maintain accurate timestamps using synchronized system clocks.
4. Protect log files from unauthorized access and modification.
5. Configure alerts for unusual authentication patterns.
6. Retain logs according to organizational security policies.
7. Use centralized log monitoring for larger environments.
8. Review successful as well as failed authentication events.

9. Results and Observations

The synthetic authentication log analysis demonstrated how security events can be identified and categorized using Linux commands.

Activity	Analysis Result
Log File Creation	Six synthetic authentication events prepared
Log Inspection	Event records reviewed
Failed Login Analysis	Four failures identified
Successful Login Analysis	Two successful logins identified
Source IP Analysis	Repeated failures from 192.0.2.10
Threat Identification	Potential password-guessing pattern observed
Security Recommendations	Monitoring, access restrictions, and authentication controls suggested

The analysis demonstrates how timestamps, usernames, source IP addresses, and authentication results help identify patterns that may require security investigation.

10. Conclusion

The Security Log Fundamentals task provided an understanding of how security logs are used to monitor system activities and identify potential threats.

Synthetic authentication logs were analyzed using Linux commands such as cat and grep to identify successful logins, failed authentication attempts, and repeated activity from specific IP addresses.

The exercise highlighted the importance of timestamps, usernames, source IP addresses, and event types in basic security investigations.

Overall, this task strengthened foundational knowledge of security event monitoring, log interpretation, and incident detection, which are essential skills in cybersecurity and Security Operations Center (SOC) environments.